

Capstone Project Proposal

University Student Competition System

Platform: Web & Mobile

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1 Introduction

The University Student Competition System (USCS) is a comprehensive web and mobile platform designed to streamline the organization, participation, and evaluation of competitions within a university. The system serves four key user roles: Students, Organizers, Judges, and Administrators, each with dedicated tools and permissions.

The platform consolidates all university competitions into one unified digital system, simplifying registration, judging, and result management. It also integrates AI-powered recommendations to suggest suitable competitions to students based on their interests and participation history, ensuring accessibility, transparency, and engagement.

2 Problem Statement

Students often miss competition opportunities due to fragmented announcements and communication.

Organizers face challenges managing registration, scheduling, and communication manually.

Judges struggle with time-consuming and inconsistent scoring methods.

Administrators lack centralized oversight of competition data and participation trends.

The absence of a unified platform limits visibility, efficiency, and fairness in competition management.

3 Project Objectives

- Develop a centralized platform to manage all competitions within the university efficiently.
- Provide role-based dashboards for students, organizers, judges, and administrators.
- Automate competition registration, evaluation, and result publication.
- Integrate AI-driven recommendations to match students with competitions that suit their skills and interests.
- Ensure transparent scoring and ranking through digital evaluation and result tracking.
- Enable real-time communication between participants, judges, and organizers.
- Ensure security, scalability, and user-friendly access across web and mobile platforms.

4 Scope of the Project

- **Multi-role access and management:** Dedicated dashboards for students, organizers, judges, and administrators, each with defined roles and permissions.
- **Competition lifecycle management:** End-to-end digital handling of event creation, registration, scheduling, evaluation, and result publishing for all competitions within the university.
- **AI-based recommendation engine:** Personalized suggestions for students based on interests, academic background, and previous participation.
- **Transparent judging and scoring:** Digital evaluation forms, automated score calculation, and real-time leaderboard updates.
- **Real-time communication and notifications:** In-app chat, alerts, and updates for announcements, schedule changes, and results.
- **Secure authentication and data privacy:** Role-based access control, encrypted data storage, and compliance with institutional security standards.
- **Scalable backend architecture:** Built using modern frameworks (e.g., Node.js/Express) and databases (e.g., MySQL/PostgreSQL) to handle large user traffic and real-time operations effectively.

5 Methodology

- **Requirement Analysis:** Conduct surveys and interviews with students, organizers, and faculty to gather detailed requirements and understand user expectations.
- **System Design:** Create architecture for the frontend, backend, and AI modules. Design data flow diagrams, ER diagrams, and user interface mockups for all user roles.
- **Development:-**
 - Frontend:** React.js for web and Flutter for mobile.
 - Backend:** Node.js/Express with a MySQL or PostgreSQL database.
 - AI Module:** Machine learning-based recommendation model for suggesting competitions.
 - Authentication:** JWT-based secure login and role-based authorization.
- **Testing:** Perform unit testing, integration testing, and user acceptance testing to ensure system reliability and ease of use.
- **Deployment:** Deploy the system to a secure cloud platform, ensuring scalability, uptime, and data protection.

6 Expected Outcomes

- A unified digital platform that manages all competitions within the university efficiently, reducing manual workload and communication gaps.
- Simplified workflows for students, organizers, judges, and administrators, ensuring clarity and collaboration.
- Automated registration, evaluation, and scoring systems that improve transparency and fairness.
- Increased student engagement through AI-powered personalized competition recommendations.
- Reliable analytics and reporting tools for administrators to track participation, performance, and trends.
- A scalable and secure digital foundation designed for future integration with other university systems such as student portals or academic management tools.